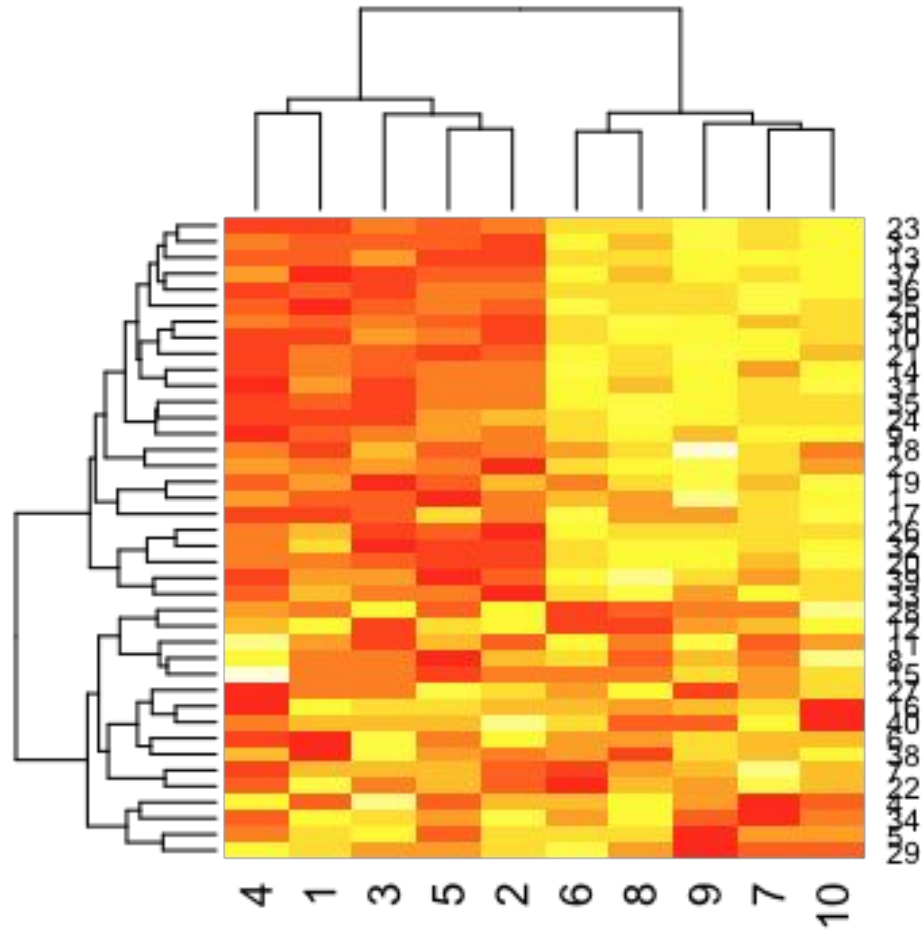


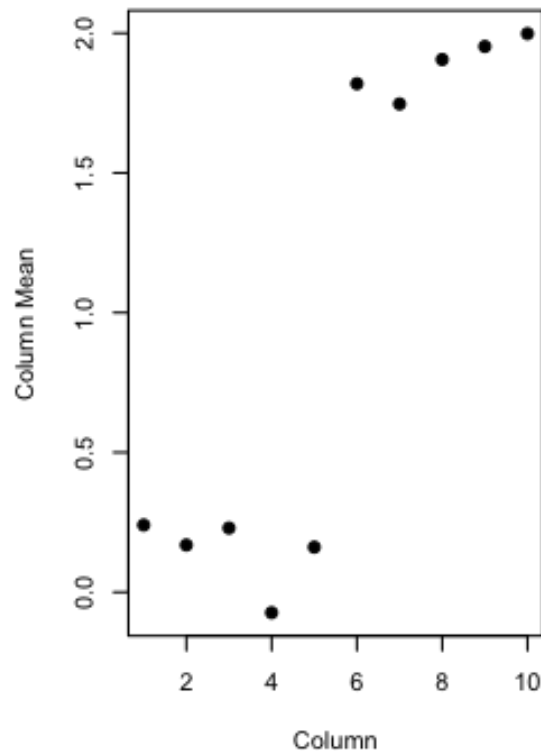
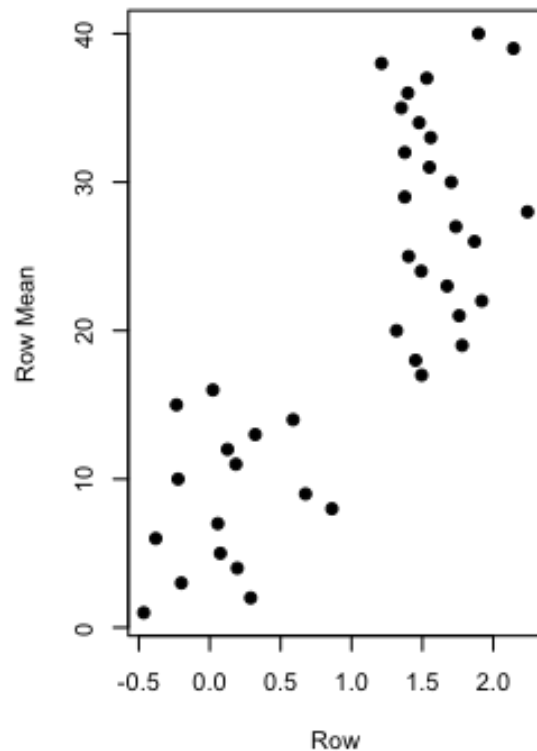
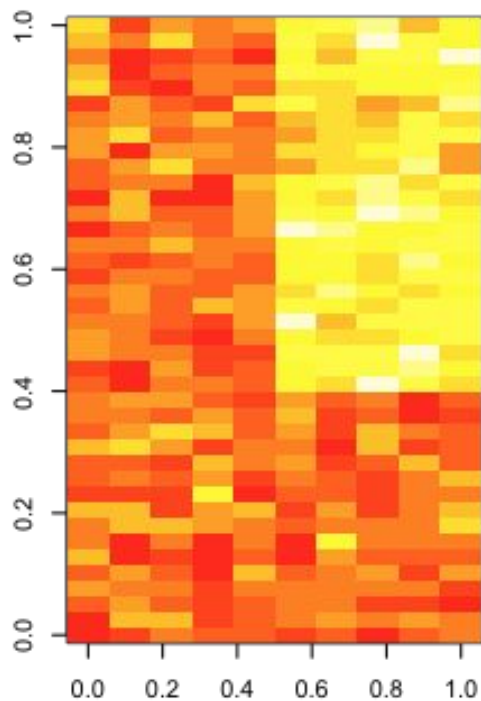
Dimension reduction

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PCA and SVD

PCA & SVD have different math goals

SVD can be used to estimate PCs

First proposed in genomics by

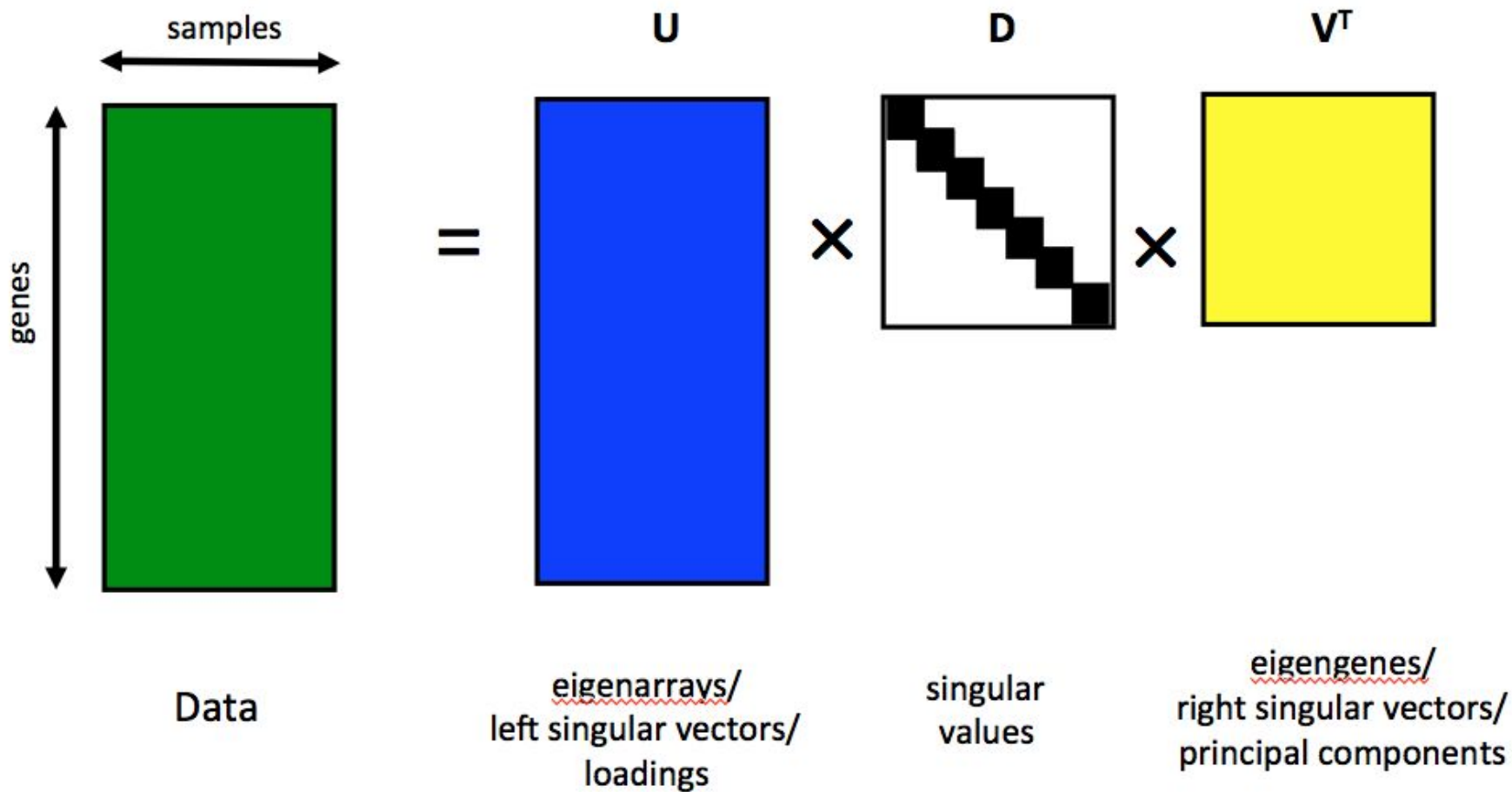
Alter et al. 2000 PNAS

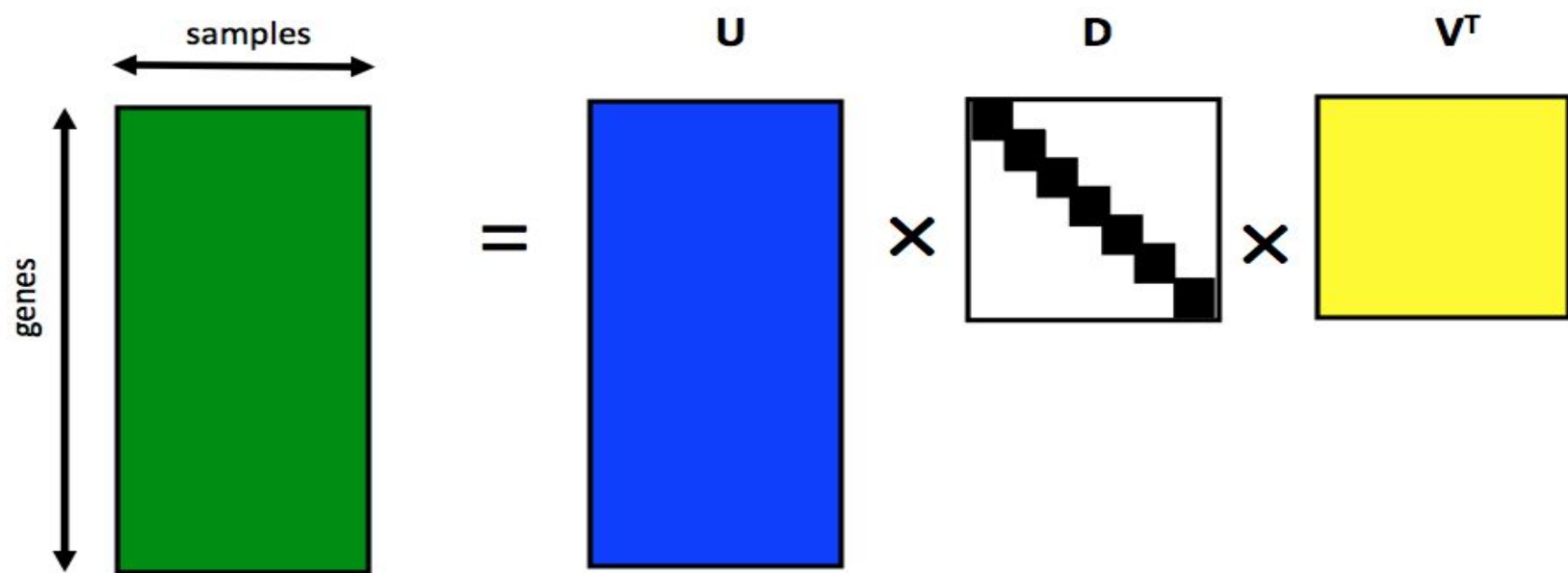
Related Problems

You have multivariate matrix of data $\mathbf{X}$

- Find a new set of multivariate variables that are uncorrelated and explain as much variance across rows as possible.
- Find the best matrix created with fewer variables (lower rank) that explains the original data.

The first goal is statistical and the second goal is data compression.

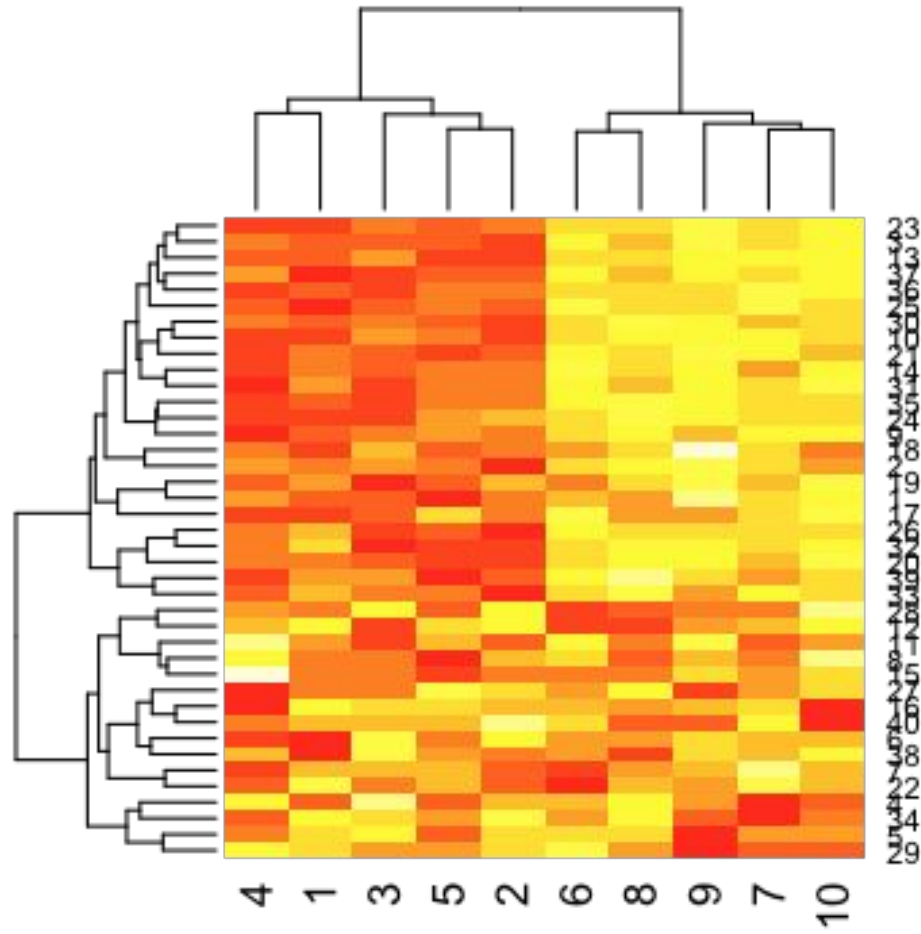


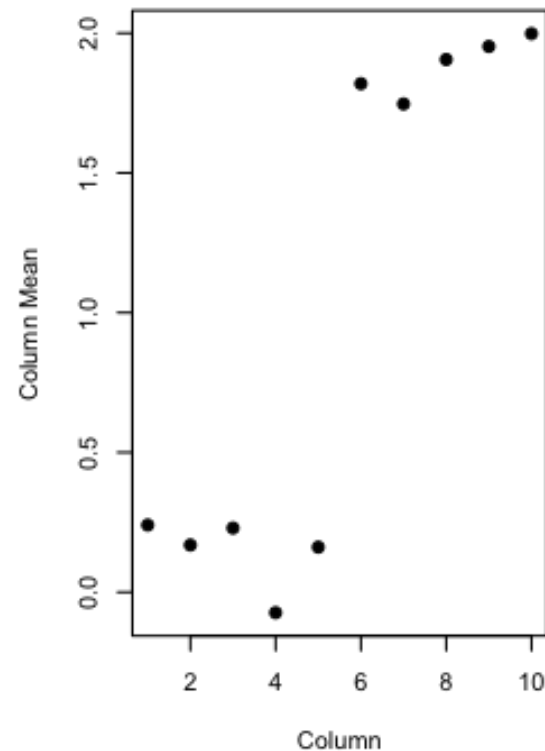
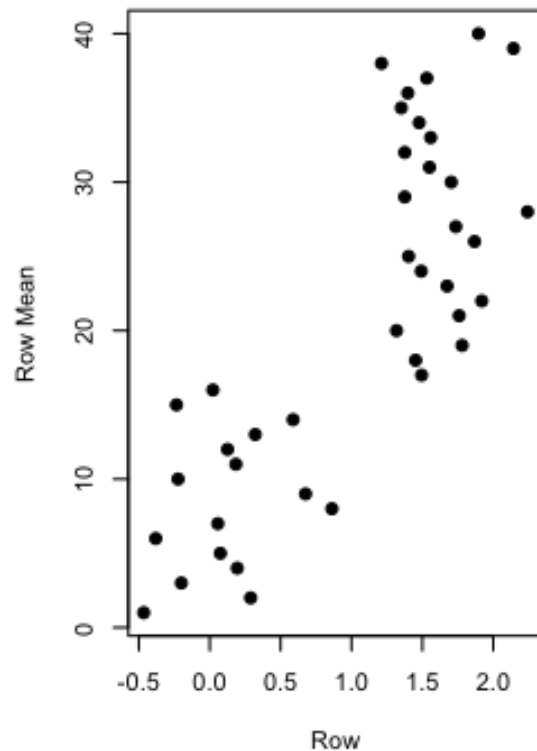
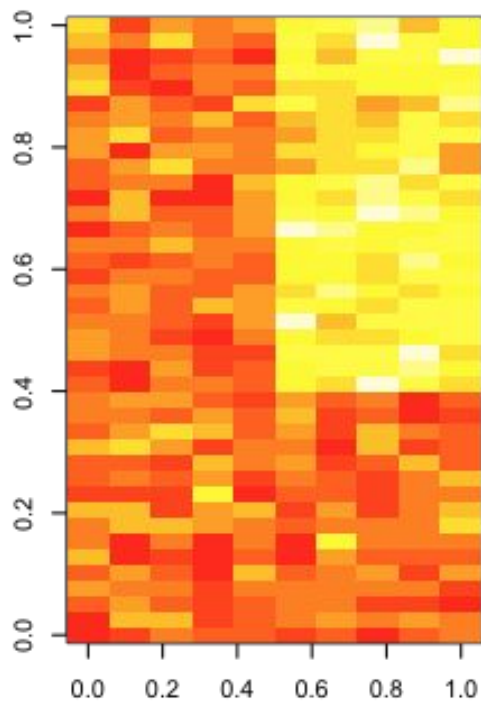


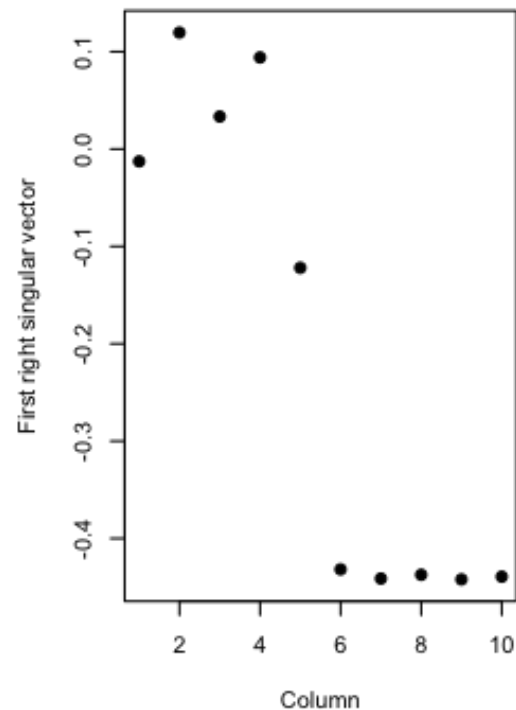
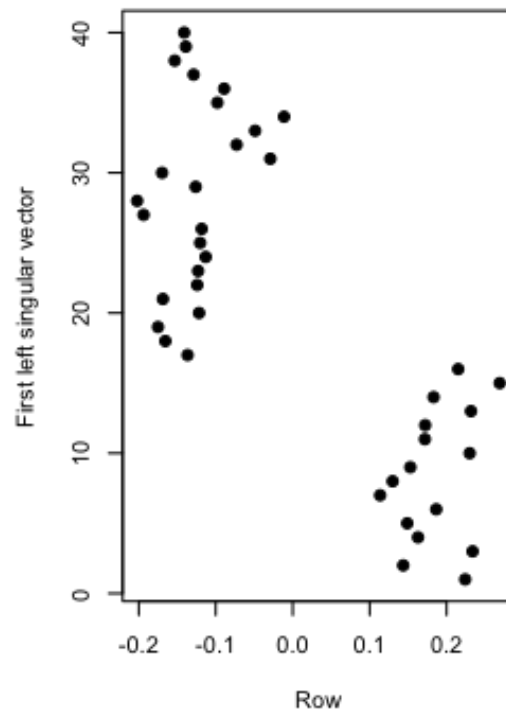
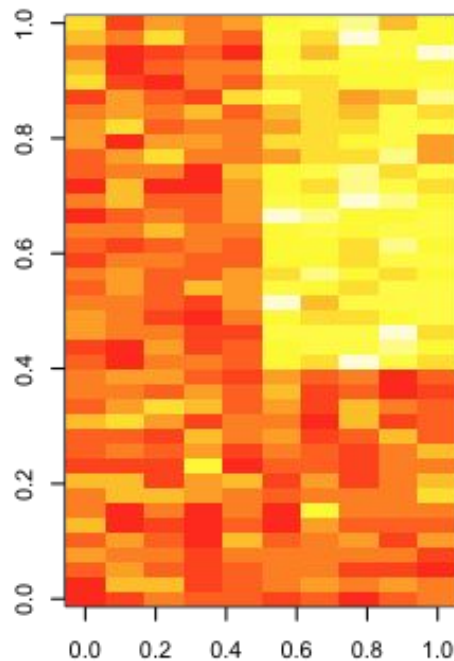
Columns of V^T /rows of U are orthogonal and calculated one at a time
 Columns of V^T describe patterns across genes
 Columns of U describe patterns across arrays

$d_i^2 / \sum_{i=1}^n d_i^2$ is the percent of variation explained by the i th column of V

Singular vectors/principal components
Method to identify patterns in the data







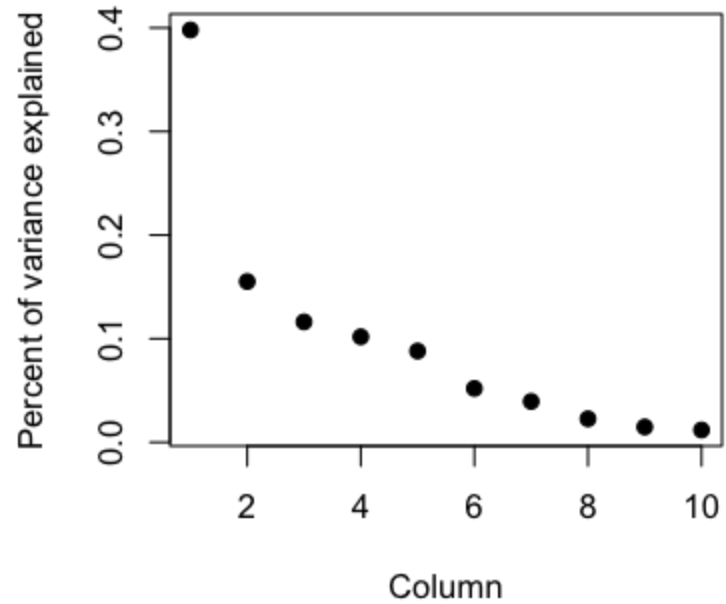
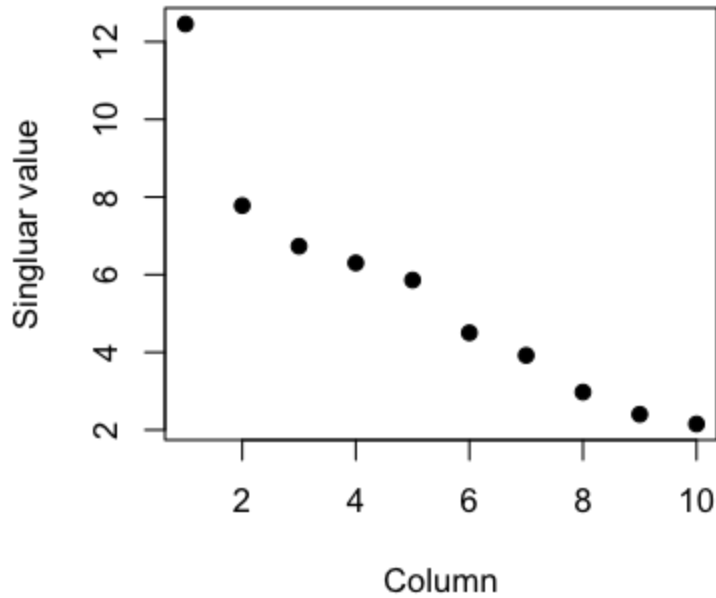
Singular values

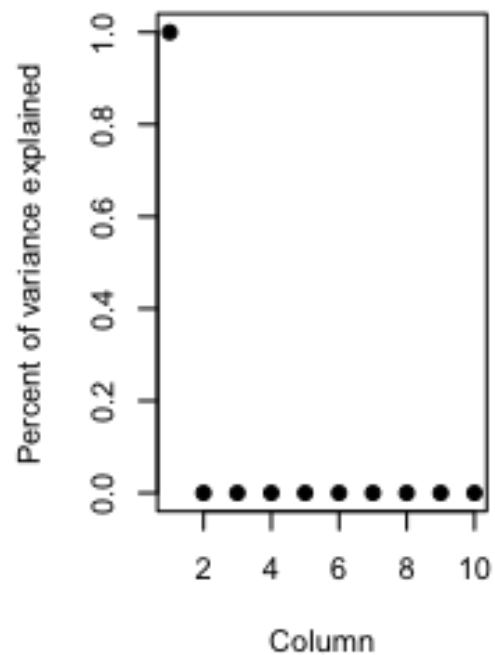
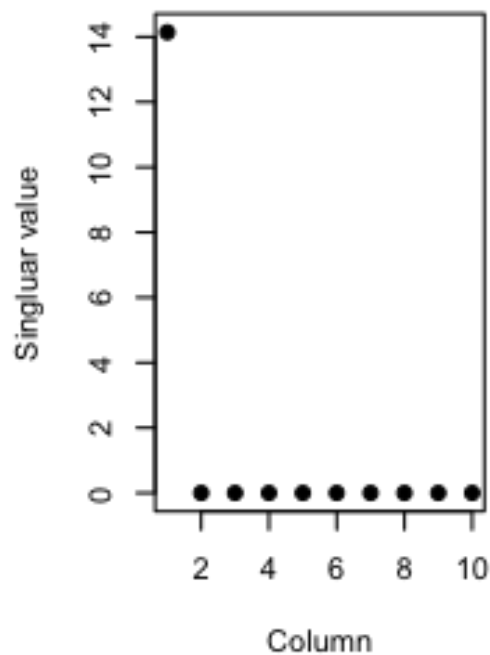
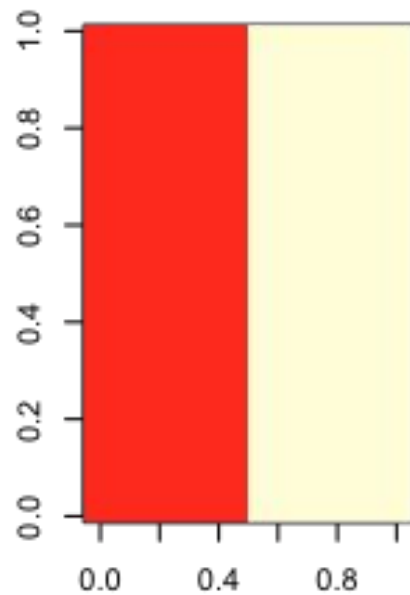
D is a diagonal matrix

d_{ii} = ith singular value

$d_{ii}^2 / \sum d_{jj}^2$ = percent variance

explained by ith singular vectors

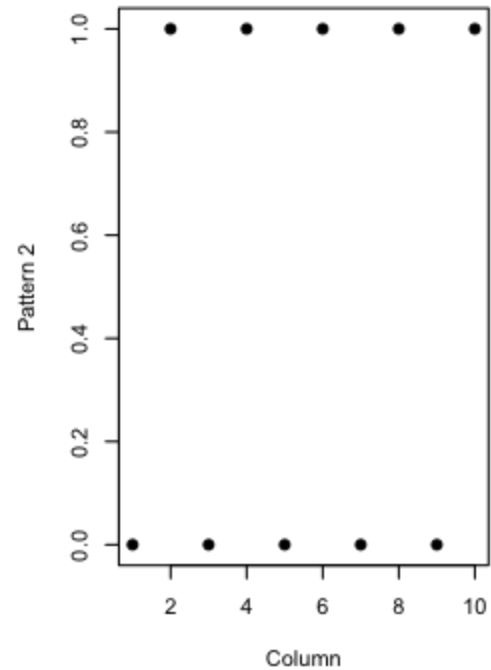
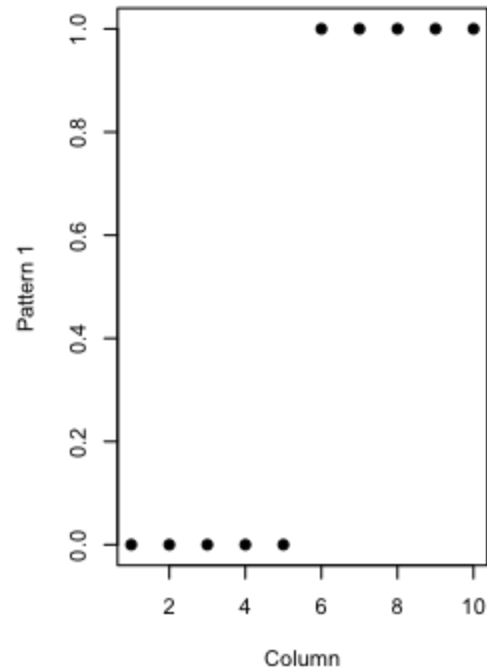
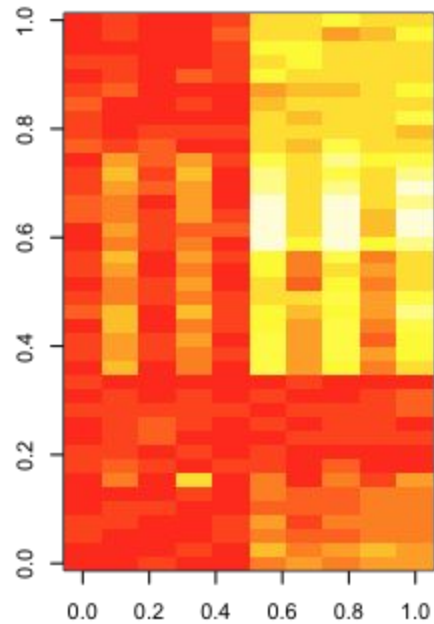


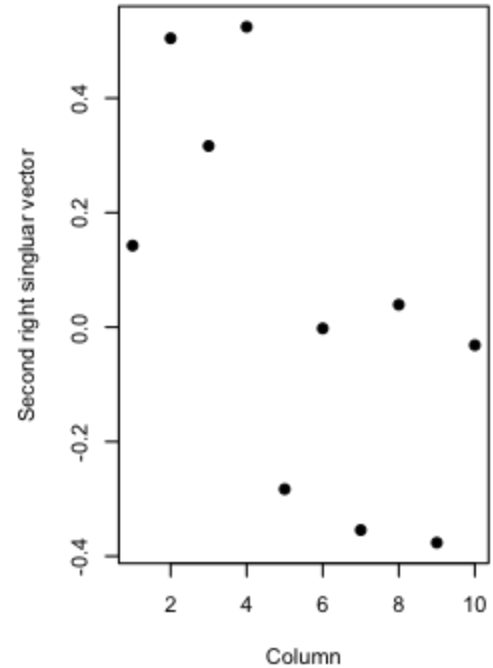
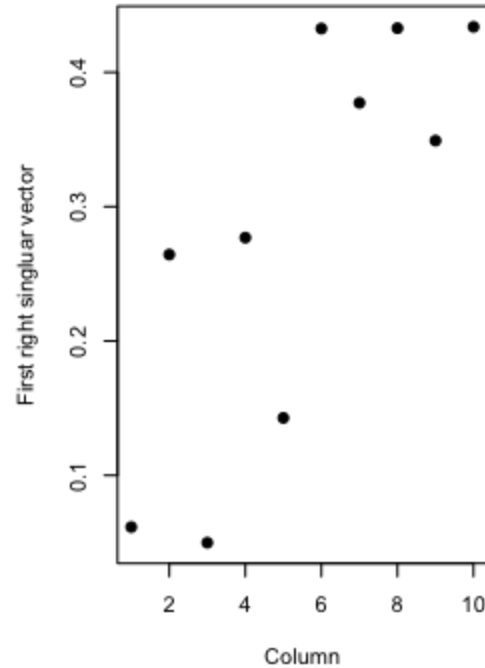
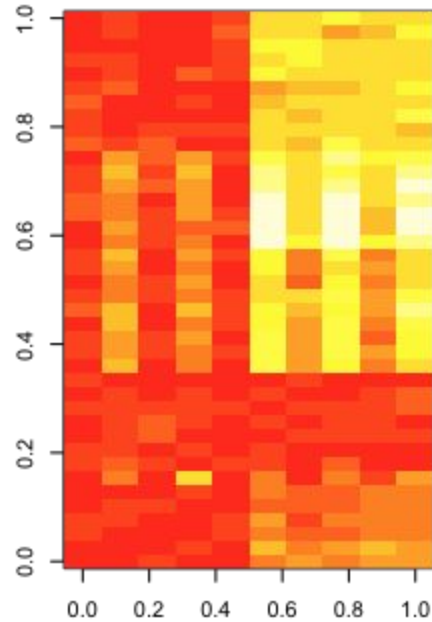


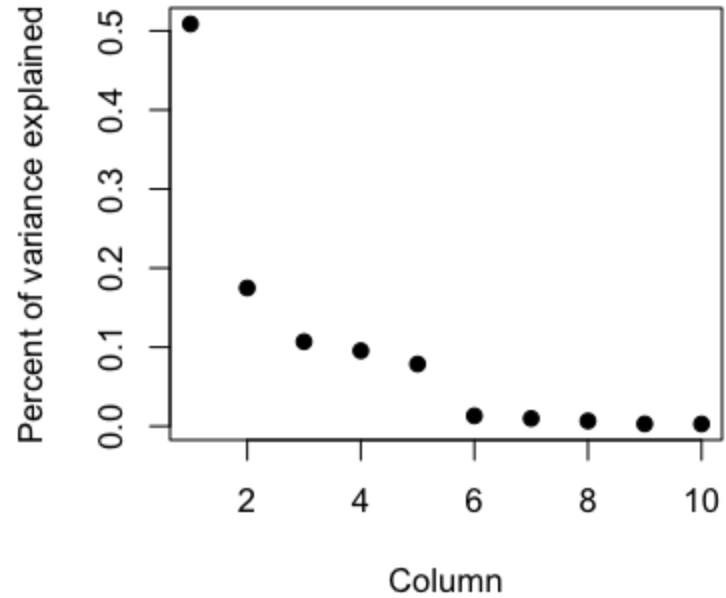
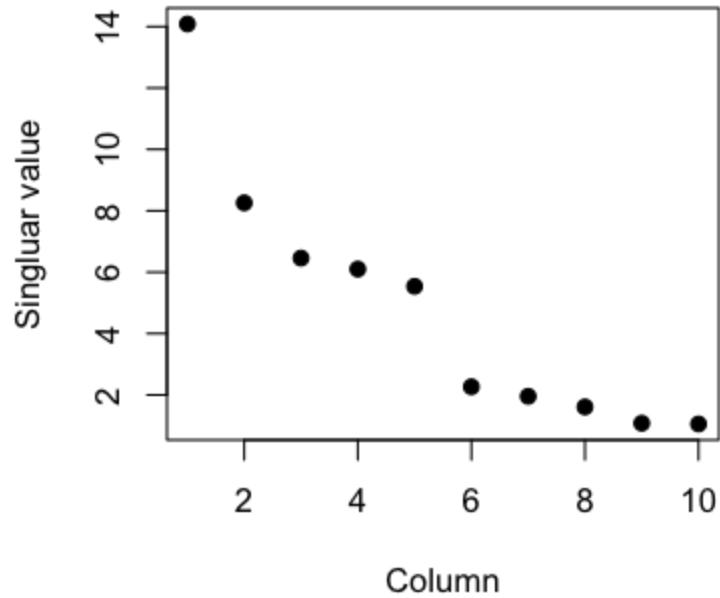
More than one pattern

Patterns are orthogonal

One PC/SV may not equal one “variable”





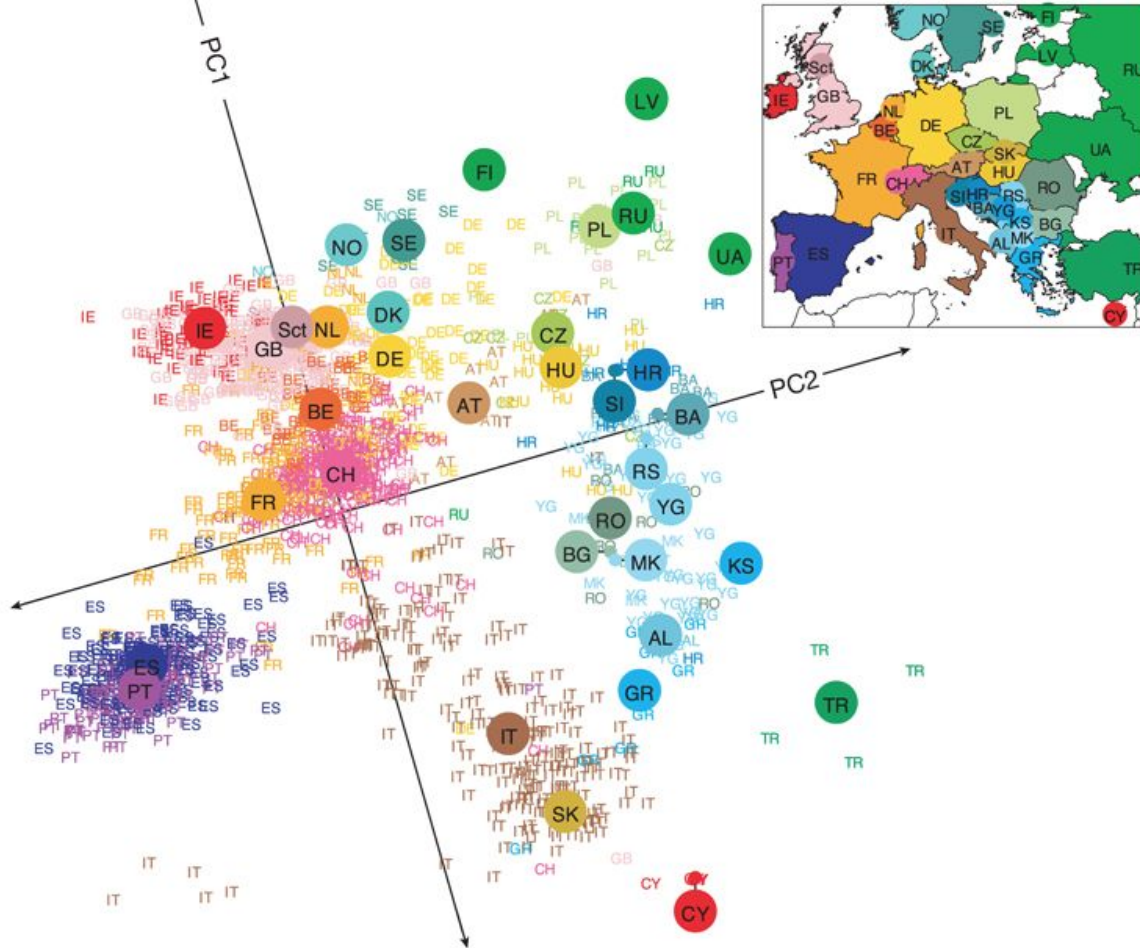


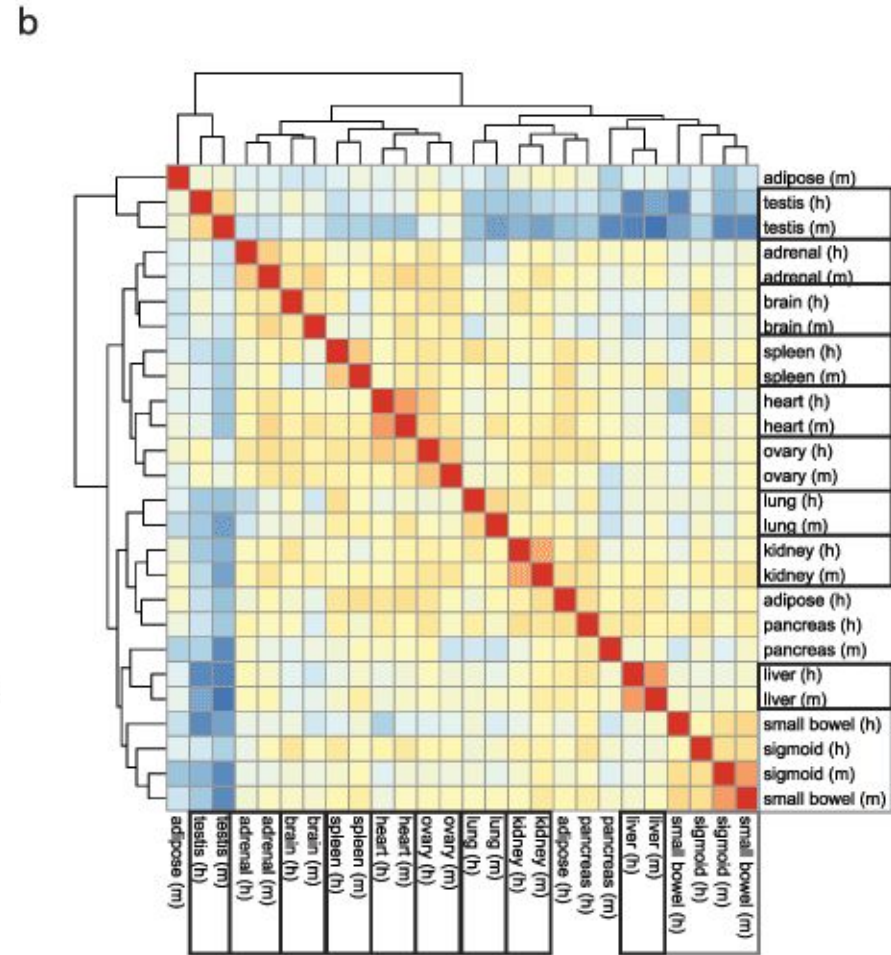
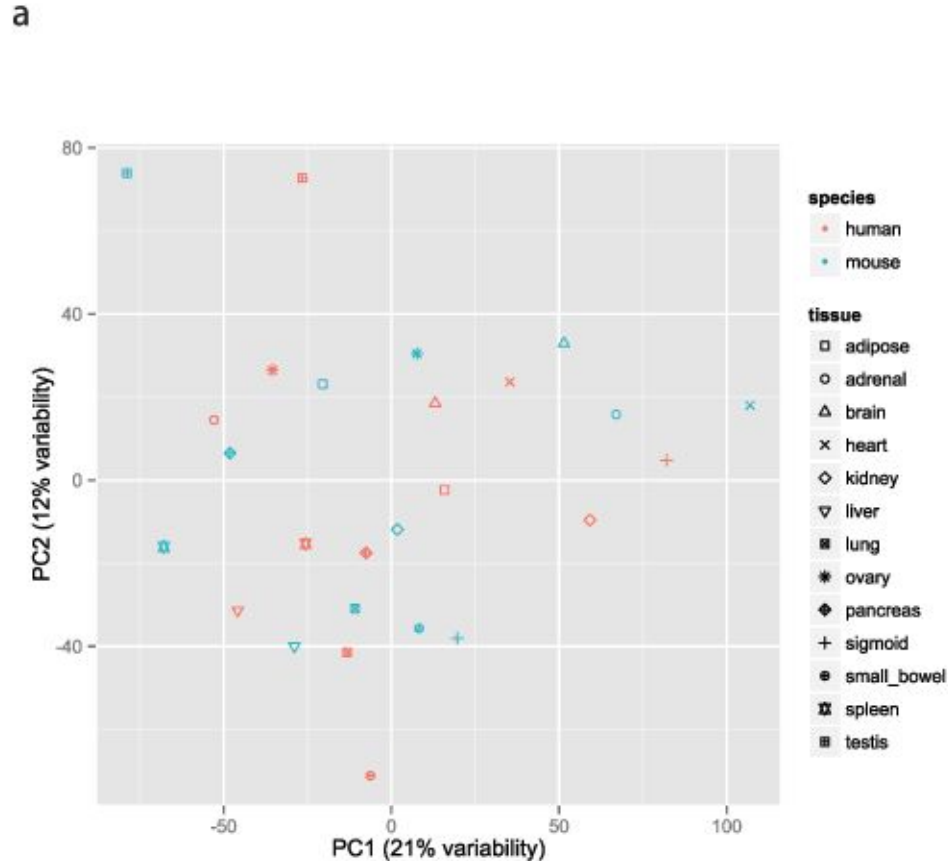
How this is used

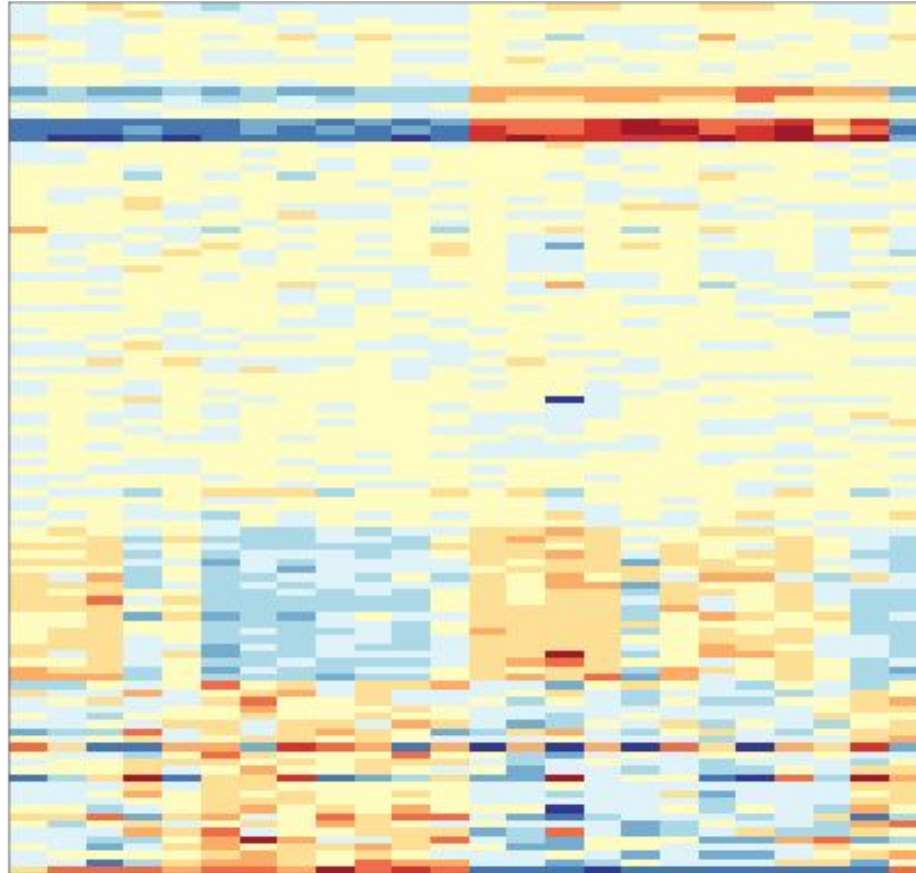
Identify meaningful patterns

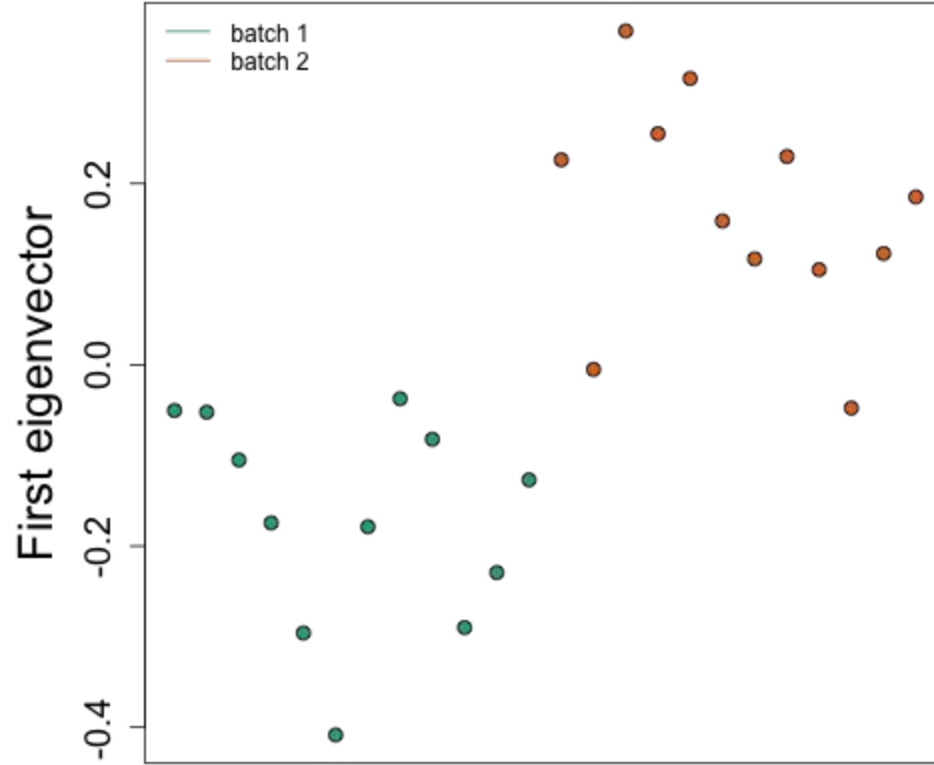
Find batch effects

PC:









Notes and further resources

- Widely used for batch effects
 - <http://www.nature.com/nrg/journal/v11/n10/full/nrg2825.html>
- There are many more decompositions people use
 - multidimensional scaling, independent components analysis, non-negative matrix factorization
- More discussion in this course
 - <https://www.edx.org/course/advanced-statistics-life-sciences-harvardx-ph525-3x>